

# EXPLORATORY DATA ANALYSIS ON



# Introduction to the Zomato Dataset

The Zomato dataset provides comprehensive information about restaurants listed on Zomato, a popular restaurant discovery and food delivery service. This dataset encompasses various aspects of restaurants, including their locations, cuisines, ratings, pricing, and operational details. The dataset can be instrumental in understanding the restaurant landscape in different regions, customer preferences, and trends in the food and beverage industry.

## **Key Features of the Dataset**

### **1. Restaurant Information**

- **name**: The name of the restaurant.
- **address**: The full address of the restaurant.
- **city**: The city where the restaurant is located.
- **locality**: The locality or neighborhood within the city.
- **longitude**: The geographical longitude coordinate of the restaurant.
- **latitude**: The geographical latitude coordinate of the restaurant.

## 2. Customer Ratings and Reviews

- aggregate\_rating**: The average rating of the restaurant given by customers, typically on a scale from 0 to 5.
- rating\_text**: The text description of the rating (e.g., 'Excellent', 'Good').
- votes**: The number of votes or reviews that contributed to the aggregate rating.

## 3. Cuisines and Dining Options

- cuisines**: The types of cuisines offered by the restaurant (e.g., Italian, Chinese, North Indian).
- average\_cost\_for\_two**: The average cost for a meal for two people at the restaurant.
- price\_range**: The price range category of the restaurant.

## 4. Operational Details

- timings**: The operational hours of the restaurant.
- highlights**: Special highlights or features of the restaurant (e.g., Live Music, Outdoor Seating).
- is\_delivery**: Indicates whether the restaurant offers delivery service.



# Characteristics

## 1.Data Types

- The dataset includes a mix of categorical (e.g., name, city, cuisines) and numerical data (e.g., aggregate\_rating, votes, average\_cost\_for\_two).

## 2.Missing Values

- Some columns, such as address and cuisines, contain missing values. Appropriate methods, such as backfilling or filling with empty strings, can handle these missing values.

## 3.Duplicates

- Duplicate records need to be checked and removed to maintain data quality and integrity.

## 4.Descriptive Statistics

- The average restaurant rating tends to be high, indicating generally positive customer experiences.
- There is a wide range of average costs for meals, catering to various customer segments from budget-friendly to high-end dining.

## 5.Popular Cuisines

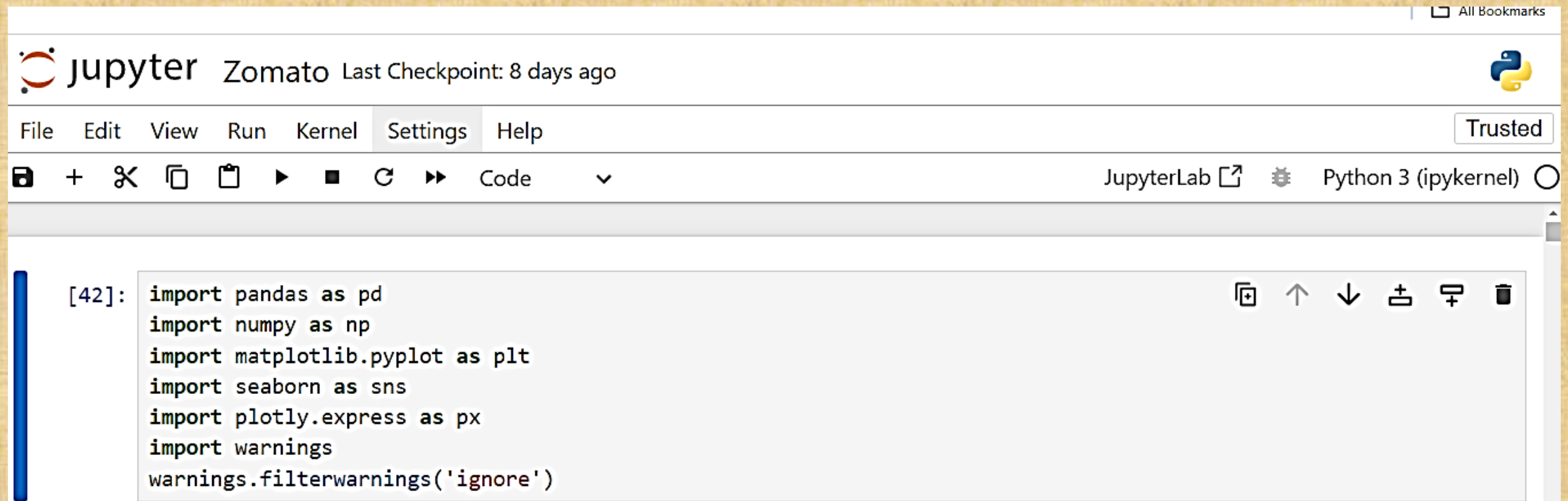
- Analysis reveals that certain cuisines, such as North Indian, Chinese, and Italian, are more commonly offered.

## 6.City-wise Distribution

- Major cities like New Delhi, Bangalore, and Mumbai have a higher concentration of restaurants.

# IMPORT LIBRARIES

To perform exploratory data analysis (EDA) on the Zomato dataset, we use a variety of Python libraries that provide tools for data manipulation, statistical analysis, and visualization. Below are the primary libraries we use in this analysis:



The screenshot shows a JupyterLab environment. The top bar includes the Jupyter logo, the name 'Zomato', and 'Last Checkpoint: 8 days ago'. A menu bar contains 'File', 'Edit', 'View', 'Run', 'Kernel', 'Settings', and 'Help'. A 'Trusted' badge is visible on the right. Below the menu bar is a toolbar with icons for file operations and execution. The main area displays a code cell with the following Python code:

```
[42]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import warnings
warnings.filterwarnings('ignore')
```

# Loading The Data-

To begin our exploratory data analysis (EDA) of the Zomato dataset, we first need to load the data into a Pandas DataFrame. This allows us to manipulate and analyze the data efficiently.

```
[2]: # Loading the data set
df = pd.read_csv('Indian-Resturants.csv')
```

```
[4]: df
```

```
[4]:
```

|   | res_id  | name        | establishment   | url                                               | address                                                 | city | city_id | locality | latitude  | longitude | ... | price_range |
|---|---------|-------------|-----------------|---------------------------------------------------|---------------------------------------------------------|------|---------|----------|-----------|-----------|-----|-------------|
| 0 | 3400299 | Bikanervala | ['Quick Bites'] | https://www.zomato.com/agra/bikanervala-khanda... | Kalyani Point, Near<br>Tulsi Cinema,<br>Bypass Road,... | Agra | 34      | Khandari | 27.211450 | 78.002381 | ... | 2           |

# Dimension of a data

```
[8]: # checking the dimensions of a data.
df.shape
```

```
[8]: (211944, 26)
```

JupyterLab   Python 3 (ipykernel)

# Checking the Data Types-

## Checking for Duplicated values-

### Checking for Duplicates:

- The df.duplicated().sum() function checks for duplicate rows in the dataset and returns the count of such rows. There are 151527 duplicated rows in a zomato dataset.
- The result is printed to show the number of duplicated rows.

### Removing Duplicates:

- The df.drop.duplicates() function removes duplicated rows from the dataset.
- The check is repeated to ensure all duplicates have been removed.

```
# checking the total duplicated values in a data.  
df.duplicated().sum()
```

```
151527
```

```
# dropping the duplicated values for data cleaning  
df = df.drop_duplicates()
```

Knowing the data types of each column is essential for understanding how the data is structured and how it can be manipulated. This also helps in identifying any potential issues, such as numerical data stored as strings.

Here is the code to check the data types:

```
# checking the data types of a data.  
df.dtypes  
  
res_id          int64  
name            object  
establishment   object  
url             object  
address         object  
city            object  
city_id         int64  
locality        object  
latitude        float64  
longitude       float64  
zipcode         object  
country_id      int64  
locality_verbose object  
cuisines         object  
timings          object  
average_cost_for_two int64  
price_range      int64  
currency         object  
highlights       object  
aggregate_rating float64  
rating_text      object  
votes            int64  
photo_count      int64  
opentable_support float64  
delivery         int64  
takeaway         int64  
dtype: object
```



# Checking Missing Values-

After removing duplicated values our next step is to find missing values in a dataset.

Missing values can significantly affect the quality of your analysis. It's essential to identify and handle them appropriately. Here is the code to check for missing values in the Zomato dataset:

# Handling Missing Values-

After applying the formula we have seen that there are 4 column where the data is missing i.e (zipcode with 47869 rows ,cuisines with 470 rows, timings with 1070 rows and opentable support with 19 rows )

Our next step is to fill those missing values with appropriate method.

1. First we will start with filling null values in categorical column with the method of mode where zipcode, cuisines and timings all are in categorical .

```
# filling the null values of a data.
```

```
df.zipcode = df.zipcode.fillna(df.zipcode.mode()[0])
```

```
df.cuisines = df.cuisines.fillna(df.cuisines.mode()[0])
```

```
df.timings = df.timings.fillna(df.timings.mode()[0])
```

```
# finding the missing values.  
df.isnull().sum()
```

|                      |       |
|----------------------|-------|
| res_id               | 0     |
| name                 | 0     |
| establishment        | 0     |
| url                  | 0     |
| address              | 18    |
| city                 | 0     |
| city_id              | 0     |
| locality             | 0     |
| latitude             | 0     |
| longitude            | 0     |
| zipcode              | 47869 |
| country_id           | 0     |
| locality_verbose     | 0     |
| cuisines             | 470   |
| timings              | 1070  |
| average_cost_for_two | 0     |
| price_range          | 0     |
| currency             | 0     |
| highlights           | 0     |
| aggregate_rating     | 0     |
| rating_text          | 0     |
| votes                | 0     |
| photo_count          | 0     |
| opentable_support    | 19    |
| delivery             | 0     |
| takeaway             | 0     |
| dtype: int64         |       |



2. In address column we will fill all null values with unknown.

```
df.address = df.address.fillna('Unknown')
```

3. As we have seen the opentable support ,country id and url column are not in use for data visualization. So we will drop this column by applying drop method.

```
# dropping the columns which is not in use.  
df.drop('opentable_support',axis=1,inplace=True)  
df.drop('url',axis=1,inplace=True)  
df.drop('country_id',axis=1,inplace=True)
```

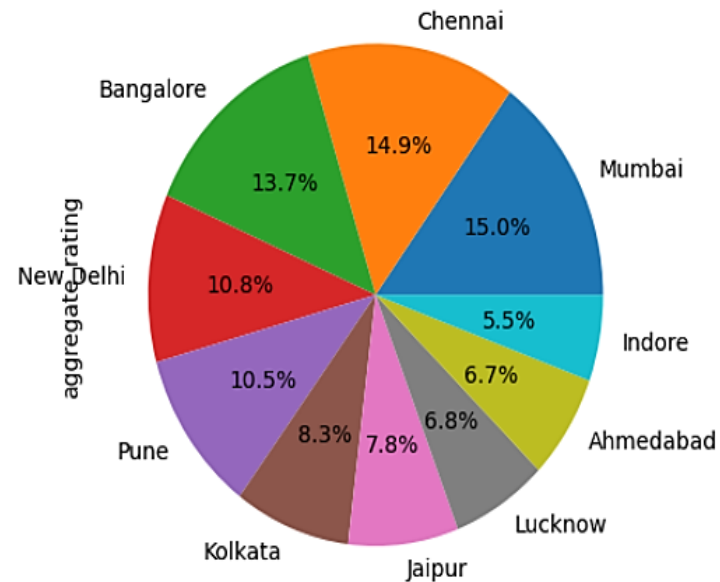
# Data visualization

## Rating Analysis

- **Pie Chart:** The pie chart displays the distribution of the total aggregate ratings among the top 10 cities. Each slice represents a city, and the size of the slice corresponds to the total aggregate rating for that city.
- **Percentage Labels:** The percentage of the total aggregate rating contributed by each city is shown on the slices, providing a clear understanding of the relative contributions of the top 10 cities.
- Where Mumbai city has the highest aggregate rating of 15.0%.
- After Mumbai, Chennai has the second highest rating of 14.9%.

```
df.groupby('city')['aggregate_rating'].sum().sort_values(ascending=False).head(10).plot(kind='pie', autopct='%1.1f%%')
```

<Axes: ylabel='aggregate\_rating'>



# Location Analysis

To get the count of restaurants in each city from the DataFrame, you can use the `value_counts()` method on the `city` column. This method returns a Series containing counts of unique values in descending order.

Here's the code to calculate and display the counts of restaurants in each city:

**In this output**

- **Chennai city has highest 2612 restaurants**
- **Mumbai city has 2538 restaurants**

```
df.city.value_counts()
```

|               |      |
|---------------|------|
| city          |      |
| Chennai       | 2612 |
| Mumbai        | 2538 |
| Bangalore     | 2365 |
| Pune          | 1911 |
| New Delhi     | 1847 |
| ...           |      |
| Udupi         | 61   |
| Howrah        | 50   |
| Neemrana      | 26   |
| Greater Noida | 22   |
| Nayagaon      | 15   |

Name: count, Length: 99, dtype: int64

## For visualization Create the Count Plot:

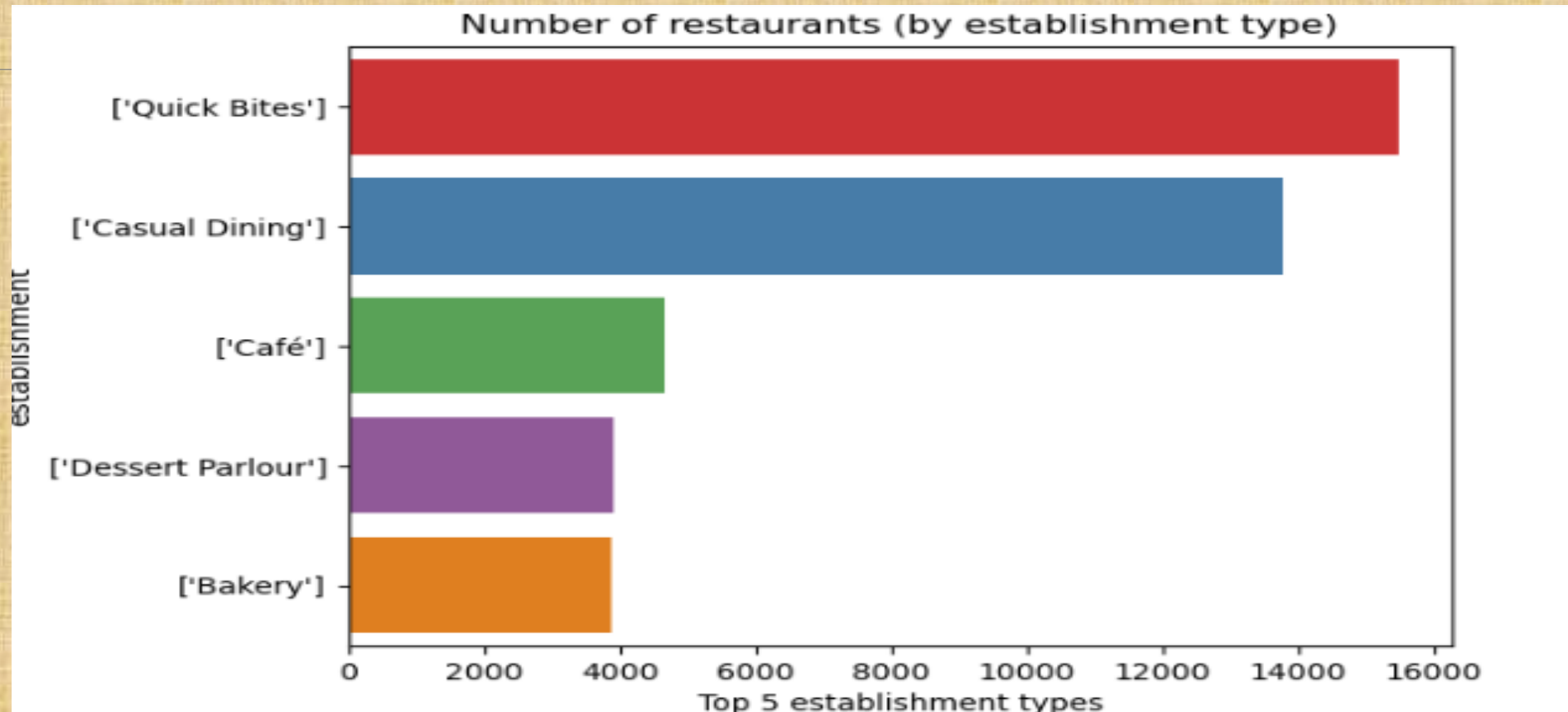
```
# Cities which have highest concentration of Restaurants
plt.figure(figsize=(14,6))
sns.countplot(x=df.city,order=df.city.value_counts().index)
plt.title('Number of Restaurants by City')
plt.xlabel('City')
plt.ylabel('Number of Restaurants')
plt.tick_params(axis='x',rotation=80)
plt.show()
```

- `sns.countplot`: This function creates a count plot, which shows the number of occurrences of each category.
- `x=df.city`: Specifies that the x-axis should represent the city column from the DataFrame.
- `order=df.city.value_counts().index`: Orders the cities by their counts in descending order. The `value_counts()` function counts the occurrences of each city, and `index` retrieves the sorted city names.



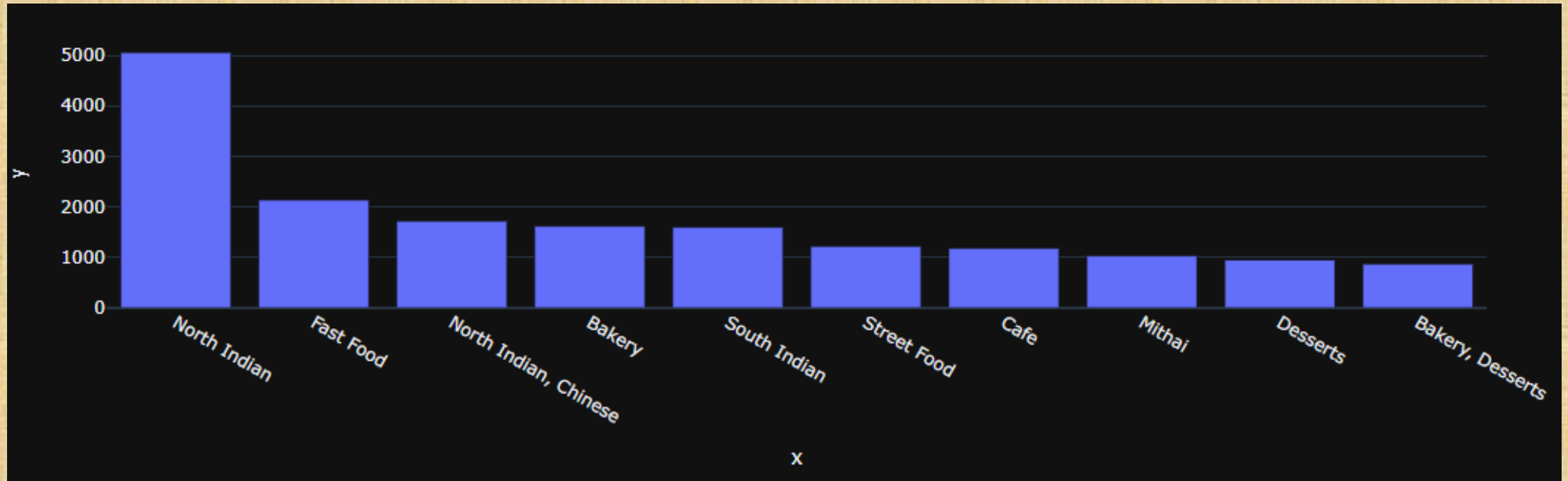


# Establishment types of no restaurants



Quick bites and casual dining type of establishment have most number of outlets

# Cuisines Analysis



In this above bar plot shows:

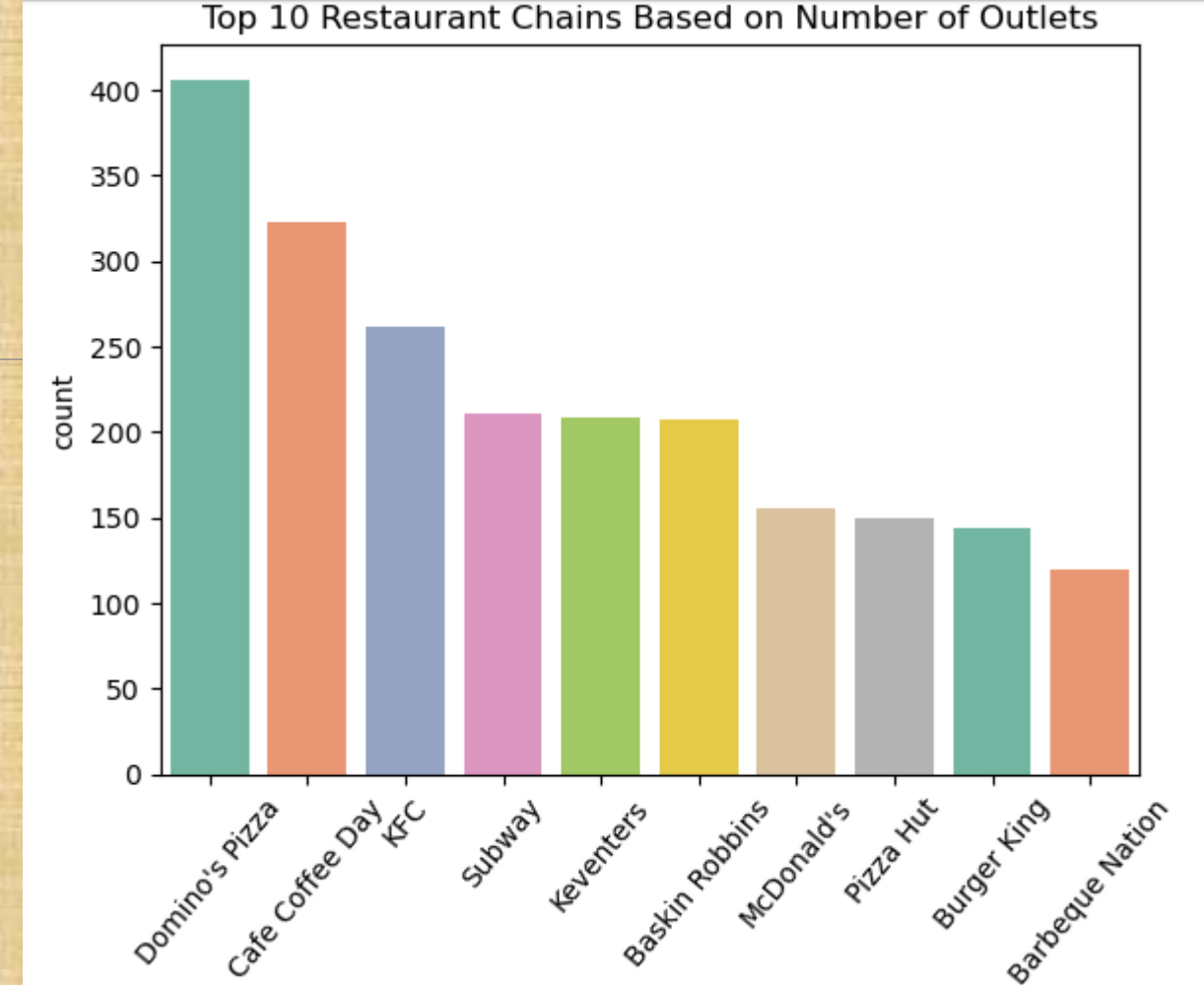
- North Indian is the most popular cuisines among buyers(5057)
- Fast Food is the second most popular cuisines (2137)

## Top Restaurant Chains:

- **Domino's Pizza** is the most widespread chain with 406 outlets, indicating its high demand and successful expansion strategy.
- **KFC** and **Café Coffee Day** follow closely, reflecting their strong market presence and brand loyalty.
- **Barbeque Nation**, with 120 outlets, rounds out the top 10, still indicating significant demand but less so than the leaders.

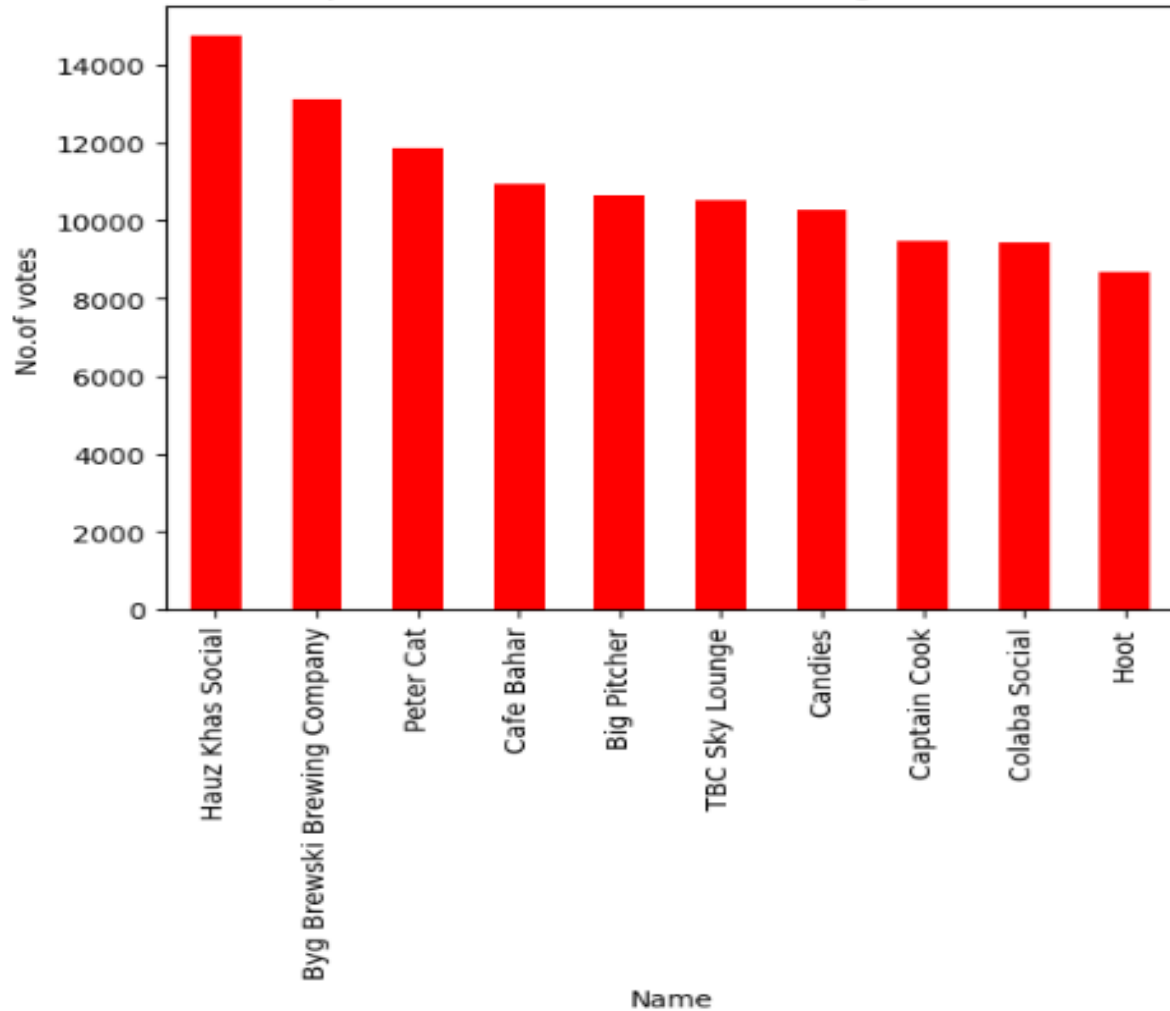
## Conclusion

The top 10 restaurant chains based on the number of outlets demonstrate high demand and successful business expansion. These chains have effectively captured market share, built strong brand recognition, and adapted well to consumer preferences. For stakeholders, such as business analysts, investors, and competitors, understanding these market leaders can provide valuable insights into successful strategies and market dynamics.



# Restaurants and votes analysis

Top 10 restaurants which have highest votes



In above barplot show top 10 restaurants which have highest average votes

- Hauz Khaz Social have the most highest average votes 14571.
- Byg Brewski Company has the second highest average votes 13114.5.
- Hoot have 8673 least average votes among top 10, still indicating significant demand but less so than the leaders.

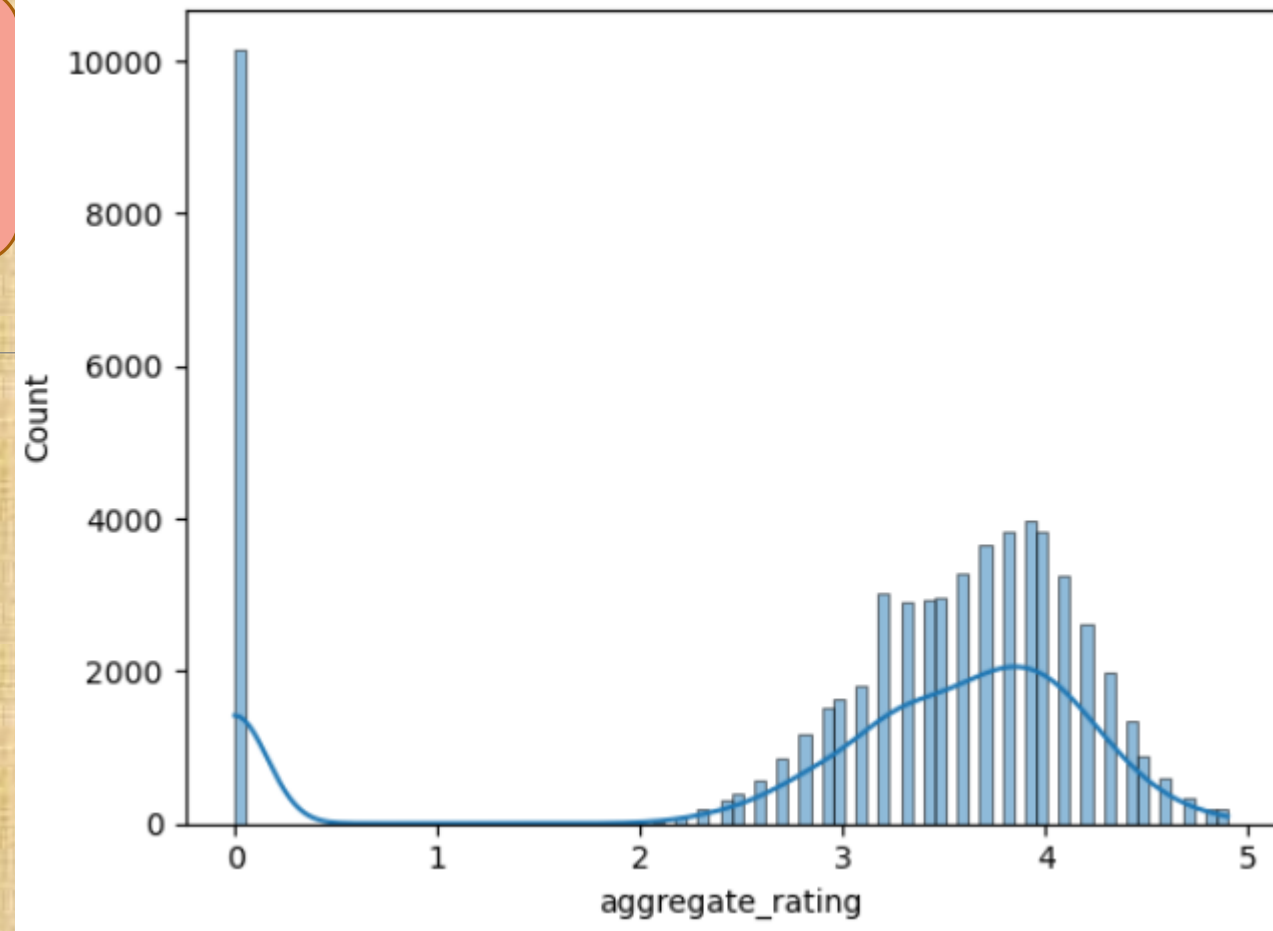


# Rating and Cost Distributions Plots

## Histogram Plot of Rating

**Negatively Skewed:** The histogram has a long tail on the left side (lower ratings) and most of the data concentrated on the right side (higher ratings).

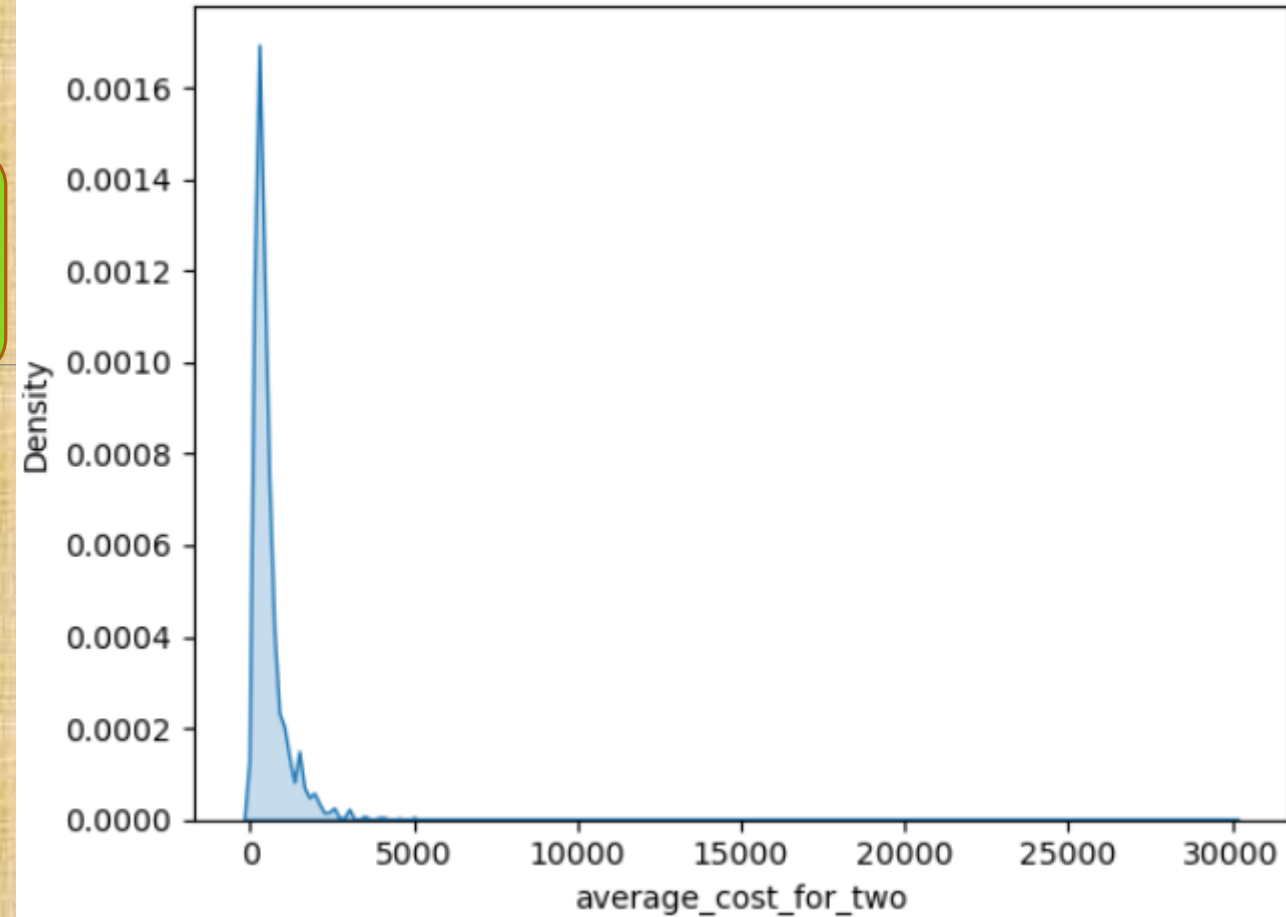
- **Central Tendency:** The most common ratings are between 3.0 and 4.0, indicating that most restaurants are rated as good to very good.
- **Spread and Variability:** The ratings range from 0 to 5, showing some variability, but the majority are clustered in the 3.0 to 4.5 range.
- **Few Low Ratings:** There are fewer restaurants with ratings below 3.0, indicating that poor-performing restaurants are in the minority.



## KDE Plot for Average Cost of Two people

With few restaurants charging average of Rs.5000+ for two, this graph is extremely skewed.

Maximum Restaurants have average cost for two between 100 to 1000.



# Data Insight between price range and aggregate ratings

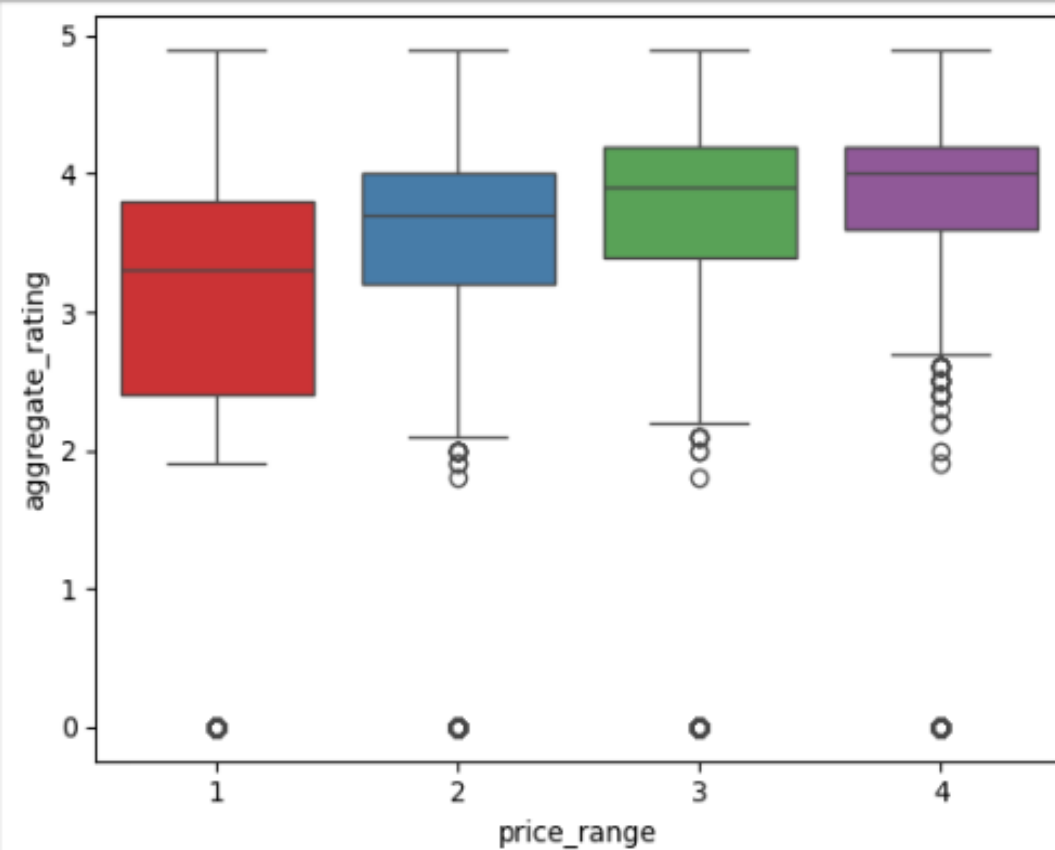
## Explanation

### 1. Visualize the Distribution of Ratings:

1. The box plot provides a visual representation of the distribution of restaurant ratings across different price ranges. Each box represents the interquartile range (IQR), with the line inside the box indicating the median rating. Whiskers extend to show the range of the data, excluding outliers.

**2. Summary Statistics:** The summary statistics provide a detailed numerical comparison of ratings within each price range.

**3. Correlation Coefficient:** The correlation coefficient provides a numerical measure of the strength and direction of the relationship between price range and restaurant ratings.



```
# Calculate summary statistics for ratings within each price range
df.groupby('price_range')['aggregate_rating'].describe()
```

|             | count   | mean     | std      | min | 25% | 50% | 75% | max |
|-------------|---------|----------|----------|-----|-----|-----|-----|-----|
| price_range |         |          |          |     |     |     |     |     |
| 1           | 30680.0 | 2.669837 | 1.568678 | 0.0 | 2.4 | 3.3 | 3.8 | 4.9 |
| 2           | 18378.0 | 3.281592 | 1.204421 | 0.0 | 3.2 | 3.7 | 4.0 | 4.9 |
| 3           | 8301.0  | 3.577653 | 1.139226 | 0.0 | 3.4 | 3.9 | 4.2 | 4.9 |
| 4           | 3058.0  | 3.701439 | 1.066155 | 0.0 | 3.6 | 4.0 | 4.2 | 4.9 |

```
# Calculate the correlation between price range and restaurant ratings
df['price_range'].corr(df['aggregate_rating'])
```

## Detailed Insights

### 1.Higher Price Ranges Tend to Have Higher Ratings:

1. Restaurants in higher price ranges (e.g., price range 4) tend to have higher average ratings compared to those in lower price ranges (e.g., price range 1). This could be due to factors such as better quality of food, ambiance, and service.

### 2.Variability in Ratings:

1. The standard deviation in ratings is generally lower for higher price ranges, indicating that ratings for higher-priced restaurants are more consistent. Lower-priced restaurants exhibit a wider range of ratings, suggesting more variability in customer satisfaction.

### 3.Outliers:

1. Outliers are present in all price ranges, indicating that there are exceptions where lower-priced restaurants receive high ratings and higher-priced restaurants receive lower ratings.

## Conclusion

Overall, the analysis indicates a positive relationship between the price range and restaurant ratings, with higher-priced restaurants generally receiving higher ratings. This insight can be useful for restaurant owners, marketers, and customers to understand the perceived value of dining experiences at different price points.



# Correlation Matrix

| correlation_matrix   |           |           |           |           |                      |             |                  |           |             |           |          |
|----------------------|-----------|-----------|-----------|-----------|----------------------|-------------|------------------|-----------|-------------|-----------|----------|
|                      | res_id    | city_id   | latitude  | longitude | average_cost_for_two | price_range | aggregate_rating | votes     | photo_count | delivery  | takeaway |
| res_id               | 1.000000  | 0.463902  | 0.009244  | -0.064495 | -0.196259            | -0.188266   | -0.253419        | -0.200272 | -0.090803   | 0.021881  | 0.0      |
| city_id              | 0.463902  | 1.000000  | 0.012329  | -0.058302 | -0.210367            | -0.207706   | -0.266710        | -0.197854 | -0.173436   | -0.036737 | 0.0      |
| latitude             | 0.009244  | 0.012329  | 1.000000  | 0.033906  | -0.002724            | -0.001703   | 0.005960         | -0.006372 | -0.002636   | 0.007394  | 0.0      |
| longitude            | -0.064495 | -0.058302 | 0.033906  | 1.000000  | -0.012011            | -0.022174   | 0.116439         | 0.023579  | 0.012057    | 0.085142  | 0.0      |
| average_cost_for_two | -0.196259 | -0.210367 | -0.002724 | -0.012011 | 1.000000             | 0.796594    | 0.255455         | 0.274503  | 0.325103    | -0.056059 | 0.0      |
| price_range          | -0.188266 | -0.207706 | -0.001703 | -0.022174 | 0.796594             | 1.000000    | 0.257688         | 0.255673  | 0.294954    | -0.052452 | 0.0      |
| aggregate_rating     | -0.253419 | -0.266710 | 0.005960  | 0.116439  | 0.255455             | 0.257688    | 1.000000         | 0.277531  | 0.229990    | 0.212771  | 0.0      |
| votes                | -0.200272 | -0.197854 | -0.006372 | 0.023579  | 0.274503             | 0.255673    | 0.277531         | 1.000000  | 0.676583    | 0.061342  | 0.0      |
| photo_count          | -0.090803 | -0.173436 | -0.002636 | 0.012057  | 0.325103             | 0.294954    | 0.229990         | 0.676583  | 1.000000    | 0.030630  | 0.0      |
| delivery             | 0.021881  | -0.036737 | 0.007394  | 0.085142  | -0.056059            | -0.052452   | 0.212771         | 0.061342  | 0.030630    | 1.000000  | 0.0      |
| takeaway             | 0.000000  | 0.000000  | 0.000000  | 0.000000  | 0.000000             | 0.000000    | 0.000000         | 0.000000  | 0.000000    | 0.000000  | 0.0      |

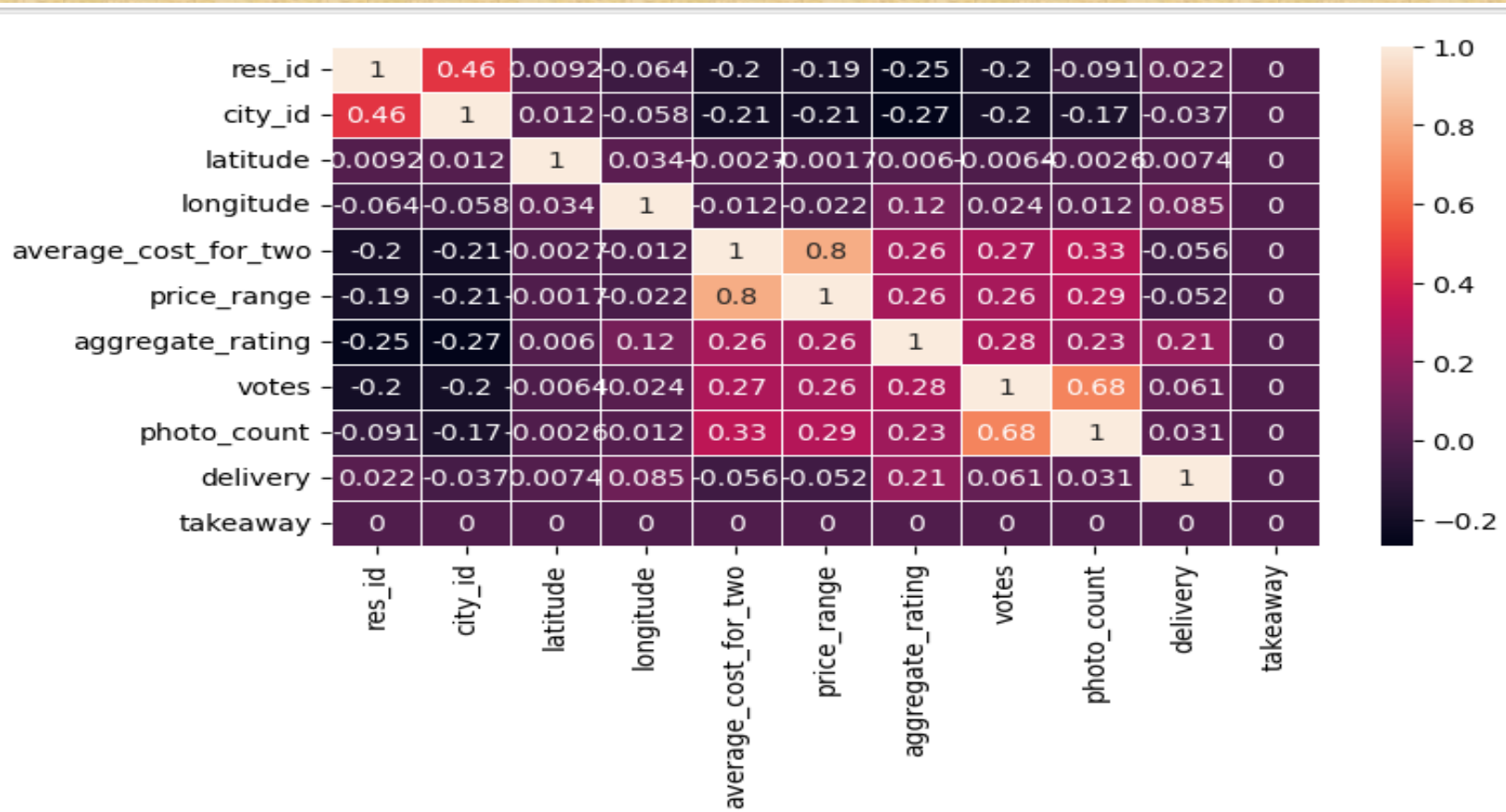
## Explanation of the Correlation Matrix

The correlation matrix shows the pairwise correlation coefficients between numeric columns in the dataset. These coefficients range from -1 to 1:

- **+1**: Perfect positive correlation, meaning that as one variable increases, the other variable increases proportionally.
- **0**: No correlation, meaning that the variables do not affect each other.
- **-1**: Perfect negative correlation, meaning that as one variable increases, the other variable decreases proportionally.

# Interpretation

- # HeatMap



# Visualization Interpretation

- **Color Intensity:** The heatmap uses color intensity to represent the strength of the correlations.
  - **Dark Red:** Strong positive correlation.
  - **Dark Blue:** Strong negative correlation.
  - **Lighter Shades:** Weaker correlations.

By examining the heatmap, you can quickly identify which pairs of variables have strong or weak correlations and the direction of these correlations (positive or negative). This visual representation helps in understanding the relationships between different numerical features in the dataset, guiding further analysis and decision-making.

## Practical Implications

- **Marketing and Business Strategy:** Understanding that higher-rated and more expensive restaurants tend to get more reviews can help businesses focus their marketing efforts on improving quality and customer satisfaction.
  - **Pricing Strategy:** Restaurants can use these insights to optimize their pricing strategies, ensuring that higher prices are justified by quality and service to maintain good ratings.
  - **Customer Experience:** Knowing the correlation between cost and ratings can help restaurants balance their pricing and service quality to attract and retain customers.
- By leveraging these insights, stakeholders can make informed decisions to enhance their restaurant's performance and customer satisfaction.



# Conclusion

The Zomato dataset provides a rich source of information for analyzing restaurant trends, customer preferences, and market dynamics. The data reveals that most restaurants are highly rated, indicating good customer satisfaction. There is a broad range of pricing, catering to various customer segments. The diversity in cuisines offered also correlates positively with higher ratings, suggesting that customers appreciate variety.

## Next Steps:

Further analysis could delve deeper into specific cities or cuisines, examining customer reviews, and exploring temporal trends to gain more insights into the restaurant industry. Additionally, predictive modeling could be applied to forecast restaurant success based on various features.



# **Thank you reading For the code of EDA**

<https://github.com/Shipra098/Zomato-Data-analysis>

